

Candidate Name: _____

Class: 07S _____



JURONG JUNIOR COLLEGE

J2 Preliminary Exam 2008

H1 Biology 8875/1

MONDAY

25 AUGUST 2008

1 HOUR

INSTRUCTIONS TO CANDIDATES:

Do NOT open this booklet until you are told to do so.

Write your name and candidate number on the answer sheet in the spaces provided.

There are thirty questions in this paper. Answer all questions. For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

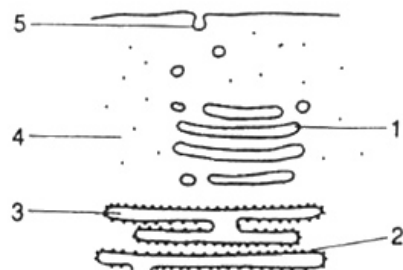
INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This Question Paper consists of 12 printed pages.

[Turn over]

- 1 Radioactive amino acids are supplied to a cell that uses them to make insulin.



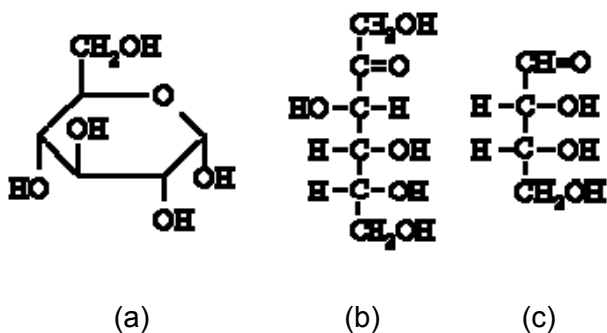
Which route will the radioactive amino acids take?

	First					Last
A	4	2	3	1	5	
B	4	3	2	1	5	
C	5	1	3	2	4	
D	5	3	2	4	1	

- 2 What is the theoretical number of chemically different dipeptides that may be assembled from 12 different amino acids?

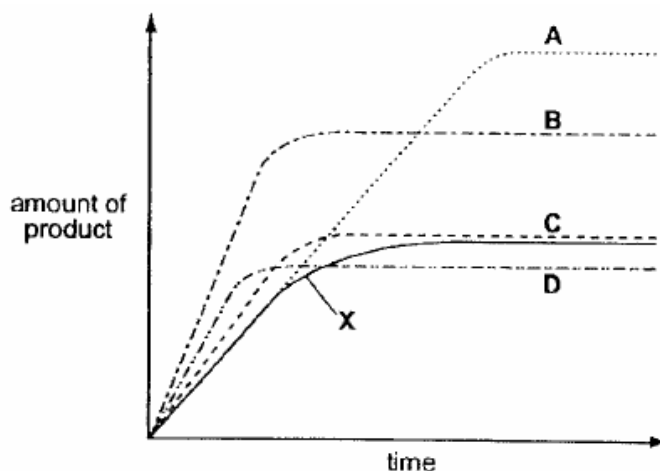
- A 24.
B 72.
C 144.
D 400.

- 3 The structures of some carbohydrates are shown in the diagram below. Which of the following statements is **true**?



- A a and c are both ketose.
B b and c are both aldose.
C a and c are both aldose.
D b and c are both ketose.

- 4 The curve X shows the activity of an enzyme at 20°C. Curves A to D show the effect of different conditions on the activity of the enzyme. Which curve shows the effect of increasing the temperature by 10°C and adding extra substrate?

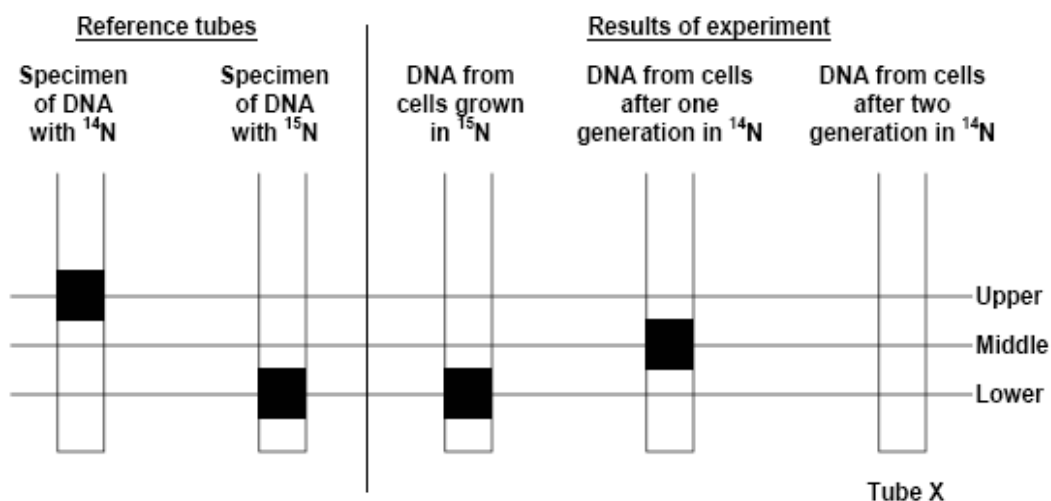


- 5 Which of the following is an example of feedback inhibition in a metabolic pathway?
- A A non competitive inhibitor binds irreversibly to the substrate.
 - B An enzyme-controlled reaction slows down as end product accumulates.
 - C There is competition between two enzymes for a common substrate.
 - D Variations in enzyme concentration affect the rate of the reaction it catalyses.
- 6 Indinavir[®] is a commercial HIV-1 protease inhibitor administered by doctors to treat AIDS patients. It is supposed to act as a competitive inhibitor. Which of the following shows the correct combination of the K_m value and maximum velocity (V_{max}) of HIV-1 protease when Indinavir[®] is added together with its substrate?

	K_m	Maximum velocity (V_{max})
A	Remains the same	Increases
B	Remains the same	Decreases
C	Increases	Remains the same
D	Decreases	Decreases

- 7 If DNA is made up of 6 different types of nitrogenous bases coding for 39 types of amino acids in proteins, what would be the minimum number of bases required for each codon?
- A 3
B 5
C 7
D 9
- 8 A cell undergoing mitosis has x picogram (pg) of DNA during prophase. What will be the amount of DNA in the same cell at the start of telophase?
- A x pg
B $x/2$ pg
C $2x$ pg
D $x/4$ pg
- 9 Cells of the bacterium *E. coli* were grown for many generations on a medium containing only the heavy isotope of nitrogen, ^{15}N . The cells were then transferred to a medium containing only ^{14}N and allowed to grow. Samples of the bacteria were removed from the culture after one generation and after two generations. The DNA from each sample was extracted and centrifuged.

The figure below shows two reference tubes and the results of this experiment.



What are the positions and relative proportions of the bands in Tube X?

	Upper	Middle	Lower
A	50 %	50 %	0 %
B	0 %	50 %	50 %
C	50 %	0 %	50 %
D	25%	50 %	25 %

- 10 You have isolated an antibiotic, edeine, which inhibits protein synthesis but has no effect on either DNA synthesis or RNA synthesis. When added to a translation mixture containing fully intact organelles, edeine stops haemoglobin translation after a short period of time, i.e. 10s. Analysis of the edeine-inhibited mixture by centrifugation showed that no polyribosomes remained by the time protein synthesis had stopped. Instead, all the mRNA accumulated, together with small ribosomal subunit and initiator tRNA.

Which step in protein synthesis does edeine inhibit?

- A It prevents the formation of the translation initiation complex, which contains the initiator tRNA and both ribosomal subunits.
 - B It inhibits the binding of aminoacyl-tRNAs to the A-site in the ribosome.
 - C It blocks the translocation of peptidyl-tRNA from the A-site to the P-site of the ribosome.
 - D It interferes with chain termination and release of the peptide.
- 11 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

normal haemoglobin

sickle cell haemoglobin

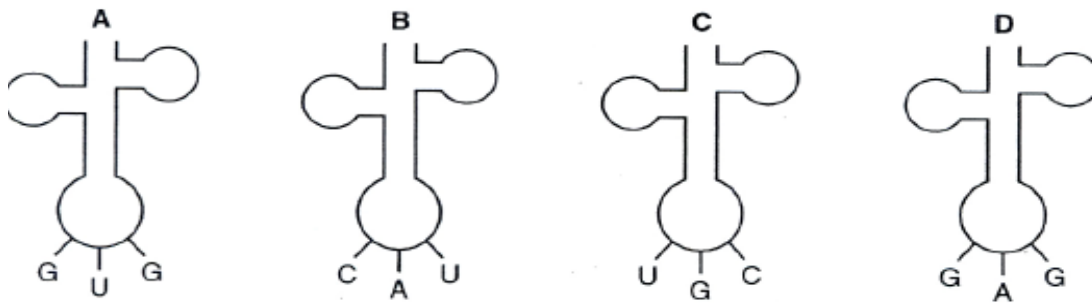
thr – pro – glu – glu

thr – pro – val – glu

mRNA codons for these amino acids are:

name of amino acid	mRNA codons	
glutamine (glu)	GAA	GAG
threonine (thr)	ACU	ACC
proline (pro)	CCU	CCC
valine (val)	GUA	GUG

Which transfer RNA molecule is involved in the formation of this part of the sickle cell haemoglobin?



12 If glucose and lactose are both absent from the growth medium, and the *lac* structural genes are expressed efficiently, what would you suspect about the condition of the *E.coli* cell?

- 1 The cell has a mutation in the operator of the *lac* operon.
- 2 The cell has a mutation in the *lacI* gene.
- 3 The cell has a mutation in the promoter of the *lac* structural gene.

- A 1 only
 B 2 only
 C 1 and 2
 D 2 and 3

13 In mice, a gene influencing coat colour has three alleles. They are

- C : black coat
 c^R : brown coat
 c : white coat

Black coat is dominant to both brown and white coat. A mouse with the genotype $c^R c$ results in another coat colour known as fawn. It is reasonable to predict that the cross

- A $C c^R \times C c^R$ would result in all black mice.
 B $c^R c^R \times c^R c$ would result in all brown mice.
 C $c^R c^R \times c c$ would result in all fawn mice.
 D $C c \times c c$ would result in all white mice.

14 A girl has blood group A and her brother has blood group B. Which combination of genotypes cannot belong to their parents?

	mother	father
A	$I^A I^A$	$I^B I^O$
B	$I^A I^B$	$I^A I^B$
C	$I^O I^O$	$I^A I^B$
D	$I^B I^O$	$I^A I^O$

15 In maize plants, two gene loci on different chromosomes have the following alleles.

Gene locus one

R : green stems

r : red stems

Gene locus two

D : normal height

d : dwarf plants

The following cross was carried out and the seeds collected.

Rr Dd x **Rr Dd**

1600 of the seeds were grown to examine the phenotypes of the offspring.

It would be reasonable to expect that among the offspring, approximately

- A** 100 offspring would be homozygous at both gene loci.
- B** 600 of the offspring will have green stems and be of normal height.
- C** $\frac{2}{3}$ of the green stem, dwarf offspring will be heterozygous at gene locus one
- D** $\frac{1}{3}$ of the red stem, normal height offspring will be heterozygous at both gene loci.

16 The haemoglobin of sufferers of the hereditary blood disorder sickle cell anaemia has only one amino acid in one of the peptide chains different to normal haemoglobin. The peptide sequence involving the differences are

Normal Haemoglobin: threonine - proline - glutamic acid - glutamic acid - lysine

Sickle cell Haemoglobin: threonine - proline - valine - glutamic acid – lysine

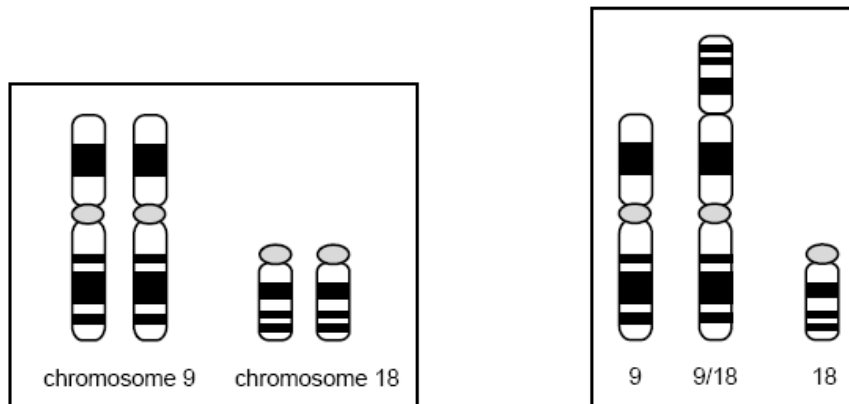
The codons for these amino acids are

Amino acid	Codons
threonine	ACU; ACC; ACA; ACG
proline	CCU; CCC; CCA; CCG
glutamic acid	GAA; GAG
valine	GUU; GUC; GUA; GUG
lysine	AAA; AAG

It is reasonable to conclude that

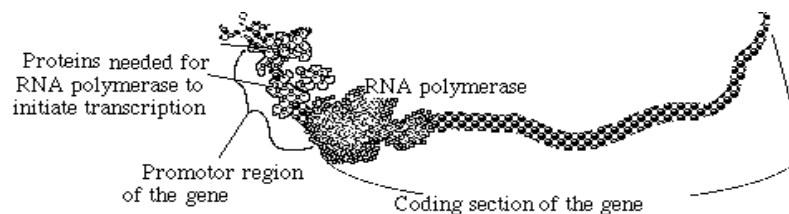
- A** the sickle cell mutation could have arisen from the eighth base in the DNA sequence of the segment being altered from adenine to uracil.
- B** the anti-codons involved in translation of the sickle cell haemoglobin could be, in order UGA – GGG – CAG – CUC – UUC.
- C** the altered sequence could have arisen during transcription, when the amino acid valine was brought into the growing peptide chain instead of the amino acid glutamic acid.
- D** the DNA sequence of the normal haemoglobin could be, in order TGA – GGG – CAG – CTT – TTT.

- 17 Genetic 'accidents' can occur to chromosomes, and are commonly known as chromosomal mutations. A scientist observed a chromosomal mutation involving chromosomes 9 and 18 in somatic cells of a male cat. The figure below shows chromosomes 9 and 18 in a normal male cat and in a cat with the chromosomal mutation.



The chromosomal mutation involved here is

- A deletion.
 - B duplication.
 - C inversion.
 - D translocation.
- 18 The ends of chromosomes that act as a "cellular clock" that control the number of times cell can divide are called:
- A centromeres
 - B telomeres
 - C spindle fibers
 - D kinases
- 19 In the illustration below the "proteins needed for RNA polymerase to initiate transcription" are:



- A mRNA molecules
- B enzymes
- C transcription factors
- D tRNA molecules

20 Which one of the following can increase the rate of transcription?

- A Binding of activators to enhancers
- B DNA demethylation
- C Histone acetylation
- D Binding of repressors to silencers

21 An inhibitor of the enzyme catalyzing the breakdown of citrate is added to respiring tissues. What will be the effect on the concentration of oxaloacetate, pyruvate and ATP?

	<i>oxaloacetate</i>	<i>pyruvate</i>	<i>ATP</i>
A	decreases	increases	decreases
B	decreases	decreases	increases
C	increases	increases	decreases
D	increases	decreases	increases

22 Which statement about respiration is **true**?

- A Cell respiration can only take place in mitochondria.
- B CO₂ is released in the conversion of glucose to pyruvate.
- C Glycolysis takes place in the mitochondrion.
- D Oxygen is the terminal electron acceptor in aerobic respiration.

23 Carbohydrates were radioactively labeled and their route from a green plant to a herbivore was identified. Which route was identified?

- | | <i>in the green plant</i> | | <i>in the herbivore</i> |
|---|----------------------------------|---|--------------------------------|
| A | glucose → starch | → | glucose → maltose → starch |
| B | glucose → starch | → | maltose → glucose → glycogen |
| C | starch → maltose | → | glucose → maltose → starch |
| D | starch → maltose | → | starch → glucose → glycogen |

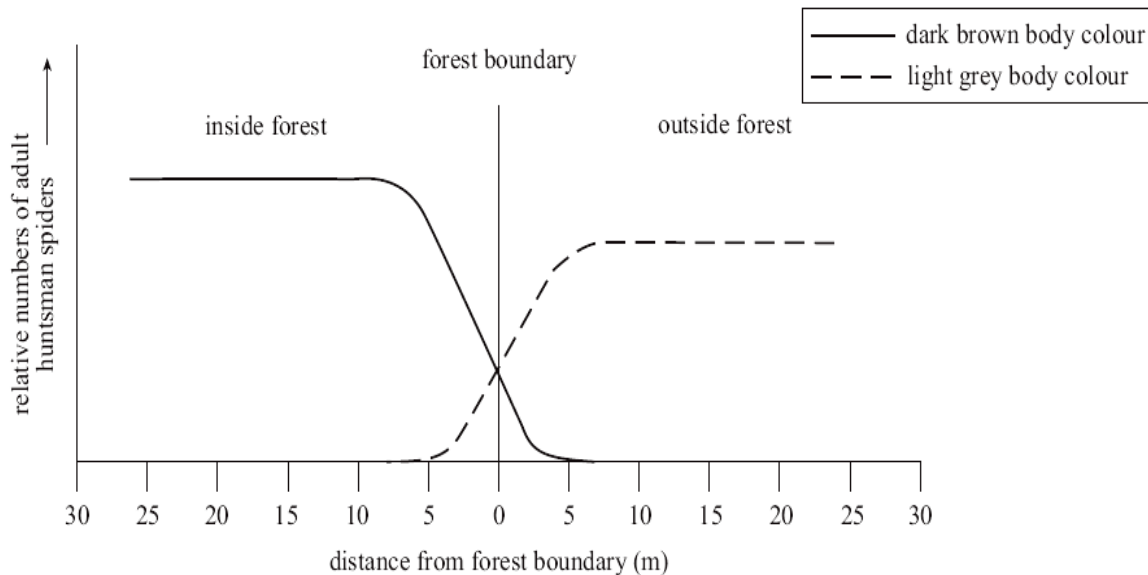
24 Which process is the link between the light dependent and the light independent stages of photosynthesis?

- A photolysis of water
- B production of carbon dioxide acceptor molecules
- C production of reduced hydrogen carrier and ATP
- D reduction of carbon dioxide

25 For allele frequencies to remain constant in a population, it is essential that

- A mutations occur
- B genetic drift occurs.
- C mating occurs at random in the population.
- D There is regular migration into the population.

26 The graph below shows the distribution of huntsman spiders at a forest boundary:

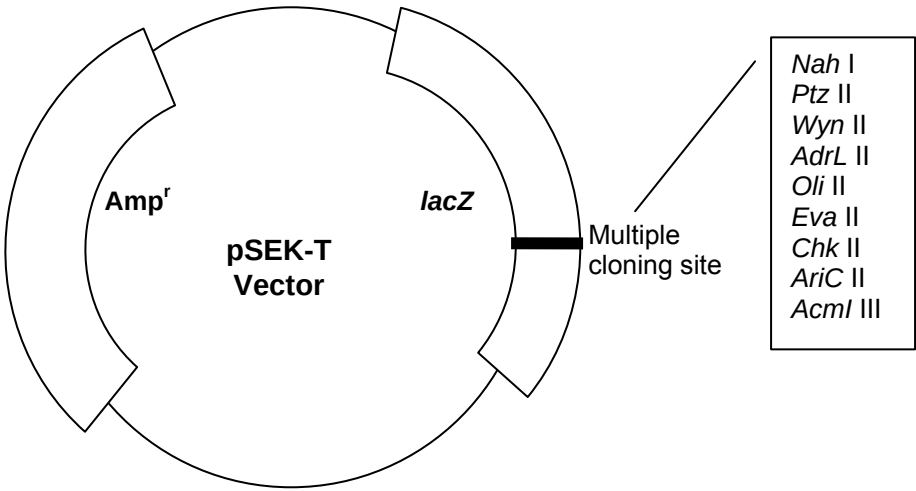


One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

The best explanation for this distribution is that

- A the two populations of spiders were once different species.
- B the two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.
- C the differences in the two habitats had changed the physical appearance of individual spiders.
- D a particular body colour provided a selective advantage to spiders in a particular habitat.

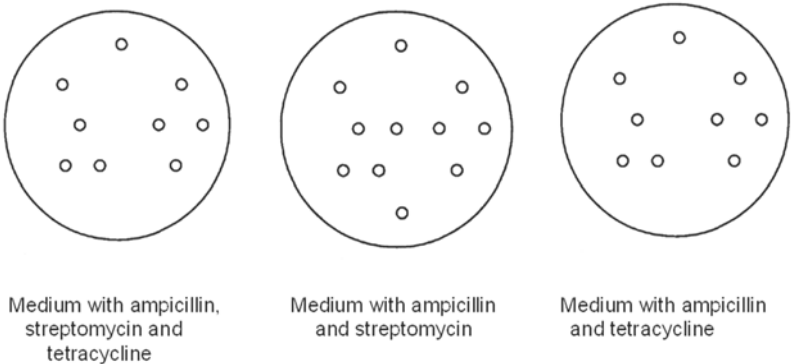
27 As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the D-ONG protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for D-ONG protein were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	<i>Chemicals to be added</i>		<i>Colour of colonies</i>
A	Ampiciliin	X-gal	Blue
B	β-galactosidase	X-gal	Blue
C	Ampiciliin	X-gal	White
D	β-galactosidase	lactose	White

28 A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coil* and the bacteria were grown on a nutrient medium with various antibiotics using replica plating. The resulting plates are shown in the diagram.



Which antibiotic gene(s) contain(s) the restriction site?

- A** Ampicillin
- B** Streptomycin
- C** Tetracycline
- D** Ampicillin and tetracycline

29 Amplification of DNA in the polymerase chain reaction requires

- A** nucleotides of uracil.
- B** DNA polymerase.
- C** primase.
- D** ribose sugar.

30 Which of the following statements is an ethical or social implication of growing Golden Rice?

- 1** There is violation of the natural organism's intrinsic values with the mixing of genes among species.
- 2** There may be objections to consuming animal genes in plants for strict vegetarians.
- 3** Emergence of resistant insects may occur.
- 4** Golden rice crops may establish themselves as superweeds.

- A** 1 only
- B** 2 and 3
- C** 1, 3 and 4
- D** 1, 2, 3 and 4

Questions	Answers	Questions	Answers
1	A	21	A
2	C	22	D
3	C	23	B
4	B	24	C
5	B	25	C
6	C	26	D
7	A	27	C
8	A	28	C
9	A	29	B
10	A	30	A
11	B		
12	C		
13	C		
14	A		
15	C		
16	B		
17	D		
18	B		
19	C		
20	A		